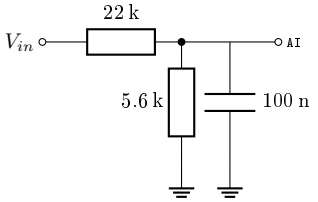
LV_ESP32 Hub — Circuitos (rango 15 V)

V_{ref} = 3.3 V (ESP32, Nano ESP32, K8) o 5 V (Uno, Nano). GND común con la placa. Un cable de la punta negra a GND.

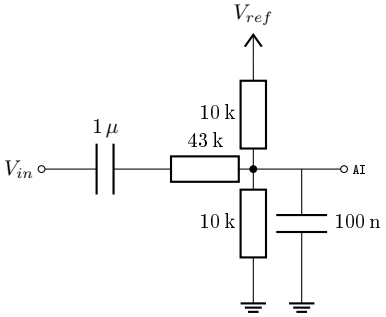
	3.3 V	5 V
V DC 0–15 V: R_1 / R_2 / sonda	22 k / 5.6 k / 4.93	22 k / 10 k / 3.2
V AC ±15 V: C / R_1 / R_b / sonda	1 µF / 43 k / 10 k / 9.6	1 µF / 27 k / 10 k / 6.4
Corriente: R_{sh} (decenas de mA / 1 A)	10 Ω / 1 Ω	10 Ω / 1 Ω
Ω: R_{ref} (1 k–100 k / decenas Ω)	10 k / 1 k	10 k / 1 k
DAC por PWM: R / C (f_c = 15.9 Hz)	10 k / 1 µF	10 k / 1 µF

1. Voltaje DC



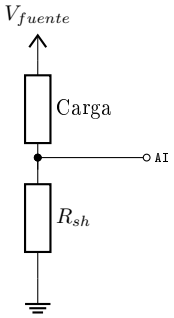
$V_{in} = V_{AI} \cdot 4.93$; AI ≤ 3.04 V a 15 V

2. Voltaje AC



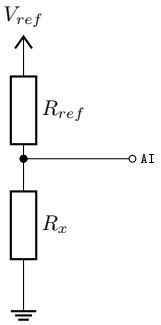
AI centrado en $V_{ref}/2$; $V_{in} = \Delta V_{AI} \cdot 9.6$

3. Corriente DC



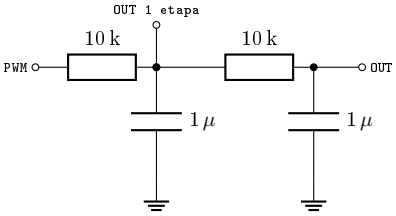
$I = V_{AI}/R_{sh}$; $V_{AI} \leq V_{ref}$

4. Resistencia y continuidad



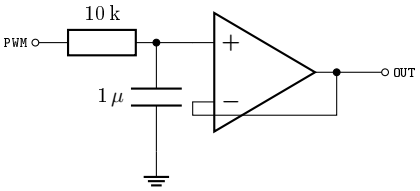
$R_x = R_{ref}\, V/(V_{ref} - V)$; continuidad: R_{ref} = 1 k, umbral 50 Ω

5. DAC por PWM (placas sin DAC)



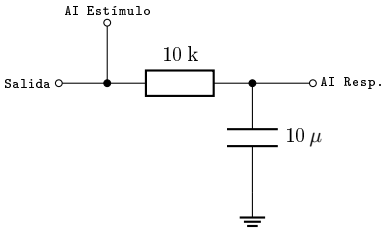
$V_{out} = V_{ref} \cdot duty$; 1 etapa f_c = 15.9 Hz, 2 etapas menos rizado

6. Salida con carga (seguidor)



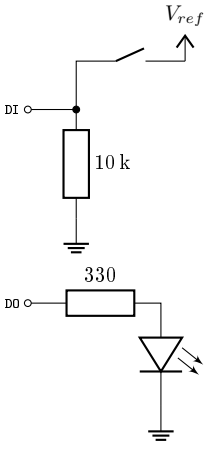
Op-amp rail-to-rail (MCP6002, LMV358) a V_{ref}

7. Bode



Circuito de prueba RC: f_c = 1.6 Hz

8. DigIn y DigOut



DI sin pull-up interno. DO: ≈ 5 mA

Protección (V DC y AC): BAT54S en el nodo AI: un diodo a V_{ref} y otro a GND. **Evitar** en ESP32: GPIO0, 2, 12, 15 con cargas que fuercen nivel al encender. Archivos LTspice: `docs/ltspice/*.asc`.